

Error No. 1

One of the problems with the Internet is that it has no real security, since it was developed as a means to share information between researchers. The protocols were developed to be robust and flexible, but security wasn't considered—it wasn't required then. It was in 1985 that the first security-related incident took place, discovered by Cliff Stoll Lawrence of Berkeley National Laboratory. When investigating a simple accounting error in the computer records of systems connected to ARPANET, Lawrence discovered that foreign governments were trying to connect to the systems in the US, and transfer data from them. Back then, most systems were located at educational and military installations.

Then in 1988, Robert T. Morris, a student at Cornell University, wrote what can be called the first Internet worm—the Morris worm. The worm, when executed from any system, connected to any available system on the network and scanned it for vulnerabilities. When it found a vulnerability, it copied itself to the system and then proceeded to propagate further—even re-infecting systems several times. The worm didn't cause any damage, but re-infected systems were slowed down—about 6,000 Unix systems were infected. Morris was convicted and sentenced to three years probation, 400 hours of community service and a fine of \$10,000. In response to the Morris worm, CERT (Computer Emergency Response Team) was established. See <http://protovision.textfiles.com/100/tr823.txt> for a full report on the incident.

The early worm gets the system			
Worm	Date discovered	Date Patch released	Method of distribution
MSBlaster	August 11, 2003	July 16, 2003	TCP port 135, 139 and 445
Welchia (uses two vulnerabilities)	August 18, 2003	March 17, 2003 July 16, 2003	TCP port 135 and 80
Swen	September 18, 2003	March, 2001	Spreads via e-mail, the KaZaA and IRC
Nimda	September 18, 2001	March 29, 2001	E-mail, Web browser, network share and known flaws in IIS Server
Code Red	July 16, 2001	June 18, 2001	Exploits flaws in IIS



Secunia provides advisories on over 2000 products

an easy search, and an online check for vulnerabilities in its test zone. The site comprehensively covers all software and hardware products—for example, it lists the HTTP Authentication Bypass Vulnerability of Axis network cameras. Just click on the products link under the Secunia advisories section on the left pane, and you will be presented with a page that has a list of all the products. Once you select a product, all the advisories for that product are listed. The Web site offers a free trial period in which you can choose to receive security warnings of a specific product in your e-mail.

Searching for information within the site is easy. All you need to do is enter 'www.secunia.com/<search string>' in your browser address bar. For example, to search for Windows buffer overflow errors, just type 'www.secunia.com/windows buffer overflow'. The test zone is located at <http://testzone.secunia.com>. Go there to automatically test

your browser, anti-virus software and operating system for vulnerabilities.

The final word

Though this may sound repetitive, there is no such thing as completely secure! If you have sensitive data such as credit card information stored on your computer, you are advised to encrypt it. Use one of the many encryption tools available online—try PGP, freely available at <http://web.mit.edu/network/pgp.html>. This is the only way to get additional security. If someone hacks your computer and manages to get your data, it can't

be read.

You need to keep in mind that patching software and keeping them up-to-date is not the only solution. As Kevin Mitnick, a former hacker, now a security consultant, said, "the weakest point in any security system is the human element". We need to ensure that best security practices are followed—there's no use in using a firewall if everyone knows your system password, or if it is easy to guess. By following proper security practices you can complete that security ring around you... well, almost.

MOULY ARUN PRABHU
mouly_arunprabhu@thinkdigit.com



CERT-CC has comprehensive security coverage



Sites to visit

A few more sites that list the latest vulnerabilities and other security news are:
www.surasoft.com
www.security-update.com
www.securityfocus.com